





EX.NO: 2 STUDY OF DIFFERENT TYPES OF NETWORK CABLES

Aim:

To study and understand different type of network cable, To make your own Ethernet cross over cable / straight cable.

a) understand different types of network cable

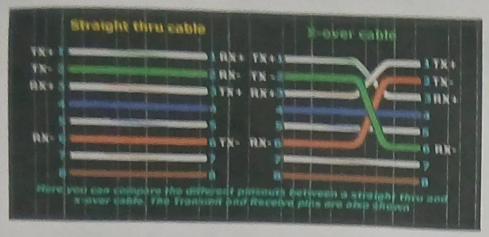
i) unshielded twisted pair (UTP) cable

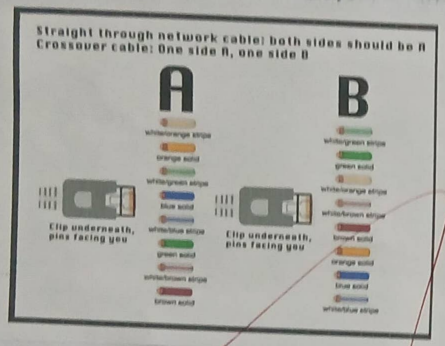
ii) shielded twisted pair (STP) cable

iii) coaxial cable

iv) fibre optic cable.

Cable Type	Category	Maximum Data Transmission	Advantages / Disadvantages	Application
UTP	category 3	10 bps	Advantages: * cheaper in cost * easy to install as they have a smaller overall diameter. Disadvantage: * More prone to (EMI)	10 Base -T Ethernet
	category 5	upto 100 Mbps		Fast Ethernet, Gigabit Ethernet.
	category 5e	1 Gbps		
STP	category 6, 6a	10 Gbps	Advantages: * shielded * Faster than UTP * less susceptible to noise and interference Disadvantages: * Expensive * Greater installation effort.	Gigabit Ethernet 10G Ethernet (55m) widely used in data centres.
SSTP	category 7	10 Gbps		Gigabit Ethernet 10G Ethernet (100m)







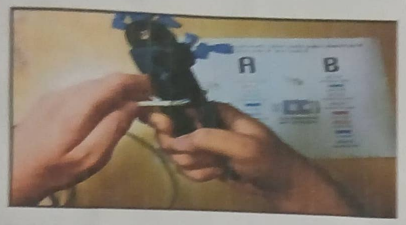
Coaxial cable	RG-6 RG-6a RG-11	10 – 100 Mbps	Advantages: * High bandwidth * Immune to interface * versatile Disadvantage: * Expensive * Require skilled * cost	speed of signal is 500m television networking High speed internet connection
Fiber optic cable	single mode multi mode	100 Gbps	Advantages: * High speed * High bandwidth Disadvantages: * Expensive * limited distance.	Maximum distance of fiber optic cable is around room.

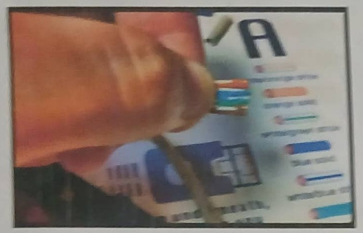
Make your own Ethernet cross-over cable / straight cable

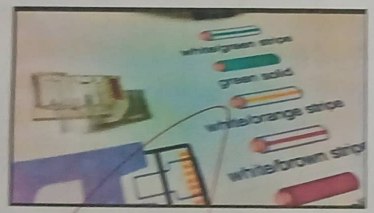
Tools and parts needed:

* Ethernet cabling. CAT 5e is certified for Gigabit support, but CAT 5 cabling could work as well, just over shorter distances.
* A crimping tool: this is an all-in-one networking tool shaped to push down the pins in the plug and strip and cut the shielding, off the cables.
* Two RJ45 plugs.
* optional two plug shields.

Difference between crossover cable and straight cable







Step 1:

To start construction of the device, begin by threading shields onto the cable.

Step 2:

Next, strip approximately 1.5 cm of cable shielding from both ends. The stripping tool has a round area to complete this task

Step 3:

After, you will need to untangle the wires; there should be four "twisted pairs" referencing back to the chart, arrange them from top to bottom. one end should be in arrangement A and the other in B.

Step 4:

Once the order is correct, bunch them together in a line, and if there are only that stick out further than others, snip them back to create an even level. the different aspect is placing these like the RJ45 plug without messing up the order.

Step 5:

Next, push the cable right in. the notch at the end of plug needs to be just over the cable shielding, and if it isn't that means that you stripped off too much shielding.

Step 6:

After the wires are securely sitting inside the plug, insert it into the crimping tool and push down.

Step 7:

Lastly, repeat for the other end using diagram B (to make a crossover cables) using diagram A (to make a straight through cables).

Result:

Thus, different types of network cables were studied successfully and straight through and cross over Ethernet cables were prepared and tested successfully.